

# Sloyd

Level 7





## About the Course

Students take folding to another level as they learn key techniques such as pleated surfaces, curved folding and crumpling. These techniques give them the tools to express their creativity and design their own models.

### Sloyd: Level 7

Level 7 starts back at the beginning of sloyd with folding techniques. However, these folding techniques are of a higher thinking level, encouraging students to envision these techniques into building structures. Paul Jackson has created videos to go along with some of the techniques, but students are encouraged to read the instructions carefully and figure these things out based on what they have learned in the previous models. While sloyd is beneficial for all students to envision geometrical forms in reality, Level 7 will be particularly interesting to the students who wish to go on to higher-level maths, engineering, design, architecture, and more.



## Placement & Combining Tips

Grade 8 students who are new to sloyd should begin with Sloyd: Level 1 and then continue courses in sequence.

Prerequisite: Sloyd: Level 6



## Scheduling

| GRADE | SCHEDULE INFO.                       | BOOKS                            |
|-------|--------------------------------------|----------------------------------|
| 8+    | Sloyd: Level 7<br>1 time/week 45 min | Folding Techniques for Designers |

### Sample Weekly View

| Day 1                 | Day 2 | Day 3          | Day 4 | Day 5 |
|-----------------------|-------|----------------|-------|-------|
| <b>Sloyd: Level 7</b> |       |                |       |       |
|                       |       | Sloyd: Level 7 |       |       |



## Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

**LINKS:** Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Sloyd: Level 7

☐ View the "Sloyd Introduction Video" in the Quick Links below for an overview of materials and resources used in Sloyd.

☐ Allow time for loading and viewing videos if available. We recommend teachers preview tutorials and work a model before teaching it.

☐ Purchase supplies and introduce students to their Sloyd tools (discuss their care and safe use). Organize them so they are easily accessible.

☐ Students will need the book and A4 paper for these models. Some models will require the paper to be cut into a square or cut in half lengthwise before beginning.

### Term Prep & Teacher Tips

Sloyd: Level 7

A4 paper (This is different from traditional, American copy paper, though it looks very similar. The measurements will be completely off if you choose to use traditional copy paper.)



## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

Sloyd: Level 7



Folding Techniques for Designers



## Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

∞ [View Supply List Details](#)

Sloyd: Level 7



Glue Sticks



Awl



Fold-and-Score Tool/Bone



A4 Paper



## Quick Links

Sloyd: Level 7

∞ [Sloyd Introduction Video](#)

Click THIS text  
or scan the QR  
code for links.



# Sloyd: Level 7

## How To Teach

|                                                                                     |                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <p><b>Overview</b></p> <p>Lessons will go much more smoothly if you prepare thoroughly. As students gain experience, develop fine motor skills, and practice habits of precision they will need less support.</p> |
|    | <p><b>Prepare</b></p> <p>Watch video tutorials or make the model yourself. Gather supplies and materials needed.</p>                                                                                              |
|    | <p><b>Recap</b></p> <p>Connect back to previous lesson and skills learned.</p>                                                                                                                                    |
|    | <p><b>Read</b></p> <p>Read the instructions one step at a time. Model each step and make sure students complete it with precision before moving to the next step.</p>                                             |
|  | <p><b>Practice</b></p> <p>When a project is complete, students may repeat it as many times as they wish to give as gifts or create original patterns similar to the project.</p>                                  |



## Term 1

### WEEK 1 45m Sloyd: Level 7 - Lesson 1

#### *Advanced Folding Techniques*

📄 Materials: Folding Techniques For Designers, A4 paper (This is different from traditional, American copy paper, though it looks very similar. The measurements will be completely off if you choose to use traditional copy paper.)

Level 7 starts back at the beginning of sloyd with folding techniques. However, these folding techniques require higher-level thinking, encouraging students to envision them as ways to build structures. Paul Jackson has created videos to go along with some of the techniques, but students are encouraged to read the instructions carefully and figure these things out based on what they have learned in the previous models. While sloyd is beneficial for all students to envision geometrical forms in reality, Level 7 will be particularly interesting to the students who wish to go on to higher-level maths, engineering, design, architecture, and more.

PREP: Students will need the book and A4 paper for these models. Some models will require the paper to be cut into a square or cut in half lengthwise before beginning.

#### → READ

Look through the diagrams on p.08 and read through the Introduction p.09-015.

#### → PRACTICE

Watch the tutorial videos.

∞ Video Link: Basic Concepts

∞ Video Link: Pleating into Sixteenths

Work through the first basic concept of Linear Divisions: Sixteenths p.016-017.

### WEEK 2 45m Sloyd: Level 7 - Lesson 2

#### *Advanced Folding Techniques*

📄 Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → PRACTICE

Watch the tutorial video.

∞ Video Link: Pleating into Thirty-Seconds

Work through Linear Thirty-Seconds p.018-019.

If you have time remaining, work through any models you have previously completed multiple times.

# Sloyd: Level 7

[Click THIS text or scan the QR code for links.](#)



## Term 1

### WEEK 3 45m Sloyd: Level 7 - Lesson 3

*Advanced Folding Techniques*

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → PRACTICE

Work through Linear Sixty-Fourths p.020.

If you have time remaining, work through any models you have previously completed multiple times.

### WEEK 4 45m Sloyd: Level 7 - Lesson 4

*Advanced Folding Techniques*

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → PRACTICE

Watch the tutorial video.

∞ Video Link: Pleating into Rotational Sixteenths

Work through Rotational Divisions: Sixteenths p.021-022.

If you have time remaining, work through any models you have previously completed multiple times.

### WEEK 5 45m Sloyd: Level 7 - Lesson 5

*Advanced Folding Techniques*

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → PRACTICE

Work through Rotational Sixteenths Variations p.023-025. You will have two lesson times to work through these models.

### WEEK 6 45m Sloyd: Level 7 - Lesson 6

*Advanced Folding Techniques*

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → PRACTICE

Work through Rotational Sixteenths Variations p.023-025. You will have two lesson times to work through these models.

# Sloyd: Level 7

[Click THIS text or scan the QR code for links.](#)



## Term 1

### WEEK 7 45m Sloyd: Level 7 - Lesson 7

*Advanced Folding Techniques*

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → PRACTICE

Watch the tutorial video.

∞ Video Link: [Pleating into Rotational Thirty-Seconds](#)

Work through Rotational Thirty-Seconds p.026.

If you have time remaining, work through any models you have previously completed multiple times.

### WEEK 8 45m Sloyd: Level 7 - Lesson 8

*Advanced Folding Techniques*

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → PRACTICE

Work through Diagonal Divisions p.027-028.

If you have time remaining, work through any models you have previously completed multiple times.

### WEEK 9 45m Sloyd: Level 7 - Lesson 9

*Advanced Folding Techniques*

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → READ

Read about grid divines on p.029

#### → PRACTICE

Work through Grid Divisions p.029-030.

These models are based on previously learned techniques. Look carefully at the diagrams to understand what you are to do.

You will have three lesson times to work through these models.

If you have time remaining, work through any models you have previously completed multiple times.

### WEEK 10 45m Sloyd: Level 7 - Lesson 10

*Advanced Folding Techniques*

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.





| Term 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <p>→ <b>PRACTICE</b></p> <p>Work through Grid Divisions p.029-030.</p> <p>These models are based on previously learned techniques. Look carefully at the diagrams to understand what you are to do.</p> <p>You will have three lesson times to work through these models.</p> <p>If you have time remaining, work through any models you have previously completed multiple times.</p>                                                                                                                                                                                                                                    |  |
| <p><b>WEEK 11</b>    <b>45m Sloyd: Level 7 - Lesson 11</b></p> <p><i>Advanced Folding Techniques</i></p> <div> <b>Materials:</b> Folding Techniques For Designers, A4 paper</div> <p>PREP: Gather and prepare supplies for this model.</p> <p>→ <b>PRACTICE</b></p> <p>Work through Grid Divisions p.029-030.</p> <p>These models are based on previously learned techniques. Look carefully at the diagrams to understand what you are to do.</p> <p>This is your last lesson to work through these models.</p> <p>If you have time remaining, work through any models you have previously completed multiple times.</p> |  |
| <p><b>WEEK 12</b>    <b>45m Sloyd: Level 7 - Lesson 12</b></p> <p><i>Exam Week</i></p> <p>→ <b>EXAMS</b></p> <p>Answer question(s) related to course.</p> <p>Questions will come from:</p> <p>Folding Techniques For Designers.</p>                                                                                                                                                                                                                                                                                                                                                                                       |  |



## Term 2

### WEEK 1 ☐ 45m Sloyd: Level 7 - Lesson 13

#### *Advanced Folding Techniques*

📄 Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → READ

Read about Symmetrical Repeats p.031. For Term 2, you will be working through the four basic types of two-dimensional symmetry: translation, reflection, rotation, and glide reflection. You will have two to four weeks to work through each of these types.

#### → PRACTICE

Watch video tutorials on symmetrical translations.

∞ Video Link: Symmetrical Motif 1

∞ Video Link: Symmetrical Motif 2

Work through the models on symmetrical translations p.031-034. You have two lessons to work through both of these models. If you finish with lesson time left, work through the models again to help further your understanding of these folds.

### WEEK 2 ☐ 45m Sloyd: Level 7 - Lesson 14

#### *Advanced Folding Techniques*

📄 Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → READ

Read about Symmetrical Repeats p.031. For Term 2, you will be working through the four basic types of two-dimensional symmetry: translation, reflection, rotation, and glide reflection. You will have two to four weeks to work through each of these types.

#### → PRACTICE

Watch video tutorials on symmetrical translations.

∞ Video Link: Symmetrical Motif 1

∞ Video Link: Symmetrical Motif 2

Work through the models on symmetrical translations p.031-034. You have two lessons to work through both of these models. If you finish with lesson time left, work through the models again to help further your understanding of these folds.

### WEEK 3 ☐ 45m Sloyd: Level 7 - Lesson 15

#### *Advanced Folding Techniques*

📄 Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

# Sloyd: Level 7

[Click THIS text or scan the QR code for links.](#)



## Term 2

### → PRACTICE

Watch video tutorials on symmetrical reflection.

∞ Video Link: Symmetrical Reflection

Work through the models on symmetrical reflection p.035. You have two lessons to work through both of these models. If you finish with lesson time left, work through the models again to help further your understanding of these folds.

### WEEK 4 45m Sloyd: Level 7 - Lesson 16

*Advanced Folding Techniques*

📄 Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

### → PRACTICE

Work through the models on symmetrical reflection p.035. You have two lessons to work through both of these models. If you finish with lesson time left, work through the models again to help further your understanding of these folds.

### WEEK 5 45m Sloyd: Level 7 - Lesson 17

*Advanced Folding Techniques*

📄 Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

### → PRACTICE

Watch video tutorials on symmetrical Rotation. Note that the video is an example of one type of rotation but is not an instructional video on how to do one of the models in the book.

∞ Video Link: Symmetrical Rotation

Work through the models on symmetrical rotation p.036-039. You have four lessons to work through all of the variation models. If you finish with lesson time left, work through the models again to help further your understanding of these folds.

### WEEK 6 45m Sloyd: Level 7 - Lesson 18

*Advanced Folding Techniques*

📄 Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

### → PRACTICE

Work through the models on symmetrical rotation p.036-039

You have four lessons to work through all of the variation models. If you finish with lesson time left, work through the models again to help further your understanding of these folds.



## Term 2

### WEEK 7 45m Sloyd: Level 7 - Lesson 19

#### *Advanced Folding Techniques*

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → PRACTICE

Work through the models on symmetrical rotation p.036-039. You have four lessons to work through all of the variation models. If you finish with lesson time left, work through the models again to help further your understanding of these folds.

### WEEK 8 45m Sloyd: Level 7 - Lesson 20

#### *Advanced Folding Techniques*

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → PRACTICE

Work through the models on symmetrical rotation p.036-039. You have four lessons to work through all of the variation models. If you finish with lesson time left, work through the models again to help further your understanding of these folds.

### WEEK 9 45m Sloyd: Level 7 - Lesson 21

#### *Advanced Folding Techniques*

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → PRACTICE

Watch video tutorials on symmetrical Glide Reflection.

∞ Video Link: [Glide Reflection](#)

Work through the model on symmetrical glide reflection p.040-043. You have three lessons to work through all of the variation models. If you finish with lesson time left, work through the models again to help further your understanding of these folds.

#### → DISCUSS

What can you build when you put multiples of these models together?  
Can you think of any buildings or structures you have seen that may be based on the glide reflection?



## Term 2

### WEEK 10 45m Sloyd: Level 7 - Lesson 22

#### *Advanced Folding Techniques*

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → PRACTICE

Work through the model on symmetrical glide reflection p.040-043. You have three lessons to work through all of the variation models. If you finish with lesson time left, work through the models again to help further your understanding of these folds.

#### → DISCUSS

What can you build when you put multiples of these models together?  
Can you think of any buildings or structures you have seen that may be based on the glide reflection?

### WEEK 11 45m Sloyd: Level 7 - Lesson 23

#### *Advanced Folding Techniques*

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → PRACTICE

Work through the model on symmetrical glide reflection p.040-043. You have three lessons to work through all of the variation models. If you finish with lesson time left, work through the models again to help further your understanding of these folds.

#### → DISCUSS

What can you build when you put multiples of these models together?  
Can you think of any buildings or structures you have seen that may be based on the glide reflection?

### WEEK 12 45m Sloyd: Level 7 - Lesson 24

#### *Exam Week*

#### → EXAMS

Answer question(s) related to course.

Questions will come from:  
Folding Techniques For Designers.



## Term 3

### WEEK 1 45m Sloyd: Level 7 - Lesson 25

#### *Advanced Folding Techniques*

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → READ

Read through Stretch & Skew on p.044.

#### → PRACTICE

Watch video tutorials on stretch & skew folds. Note that the video is not in English. The purpose is to see an example worked out.

∞ Video Link: Stretch & Skew

Work through the models on stretch & skew p.044-049. You have five lessons to work through all of the variation models. If you finish with lesson time left, work through the models again to help further your understanding of these folds.

#### → DISCUSS

What can you build when you put multiples of these models together?  
Can you think of any buildings or structures you have seen that may be based on this fold?

### WEEK 2 45m Sloyd: Level 7 - Lesson 26

#### *Advanced Folding Techniques*

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → PRACTICE

Work through the models on stretch & skew p.044-049. You have five lessons to work through all of the variation models. If you finish with lesson time left, work through the models again to help further your understanding of these folds.

#### → DISCUSS

What can you build when you put multiples of these models together?  
Can you think of any buildings or structures you have seen that may be based on this fold?

### WEEK 3 45m Sloyd: Level 7 - Lesson 27

#### *Advanced Folding Techniques*

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → PRACTICE

Work through the models on stretch & skew p.044-049. You have five



## Term 3

lessons to work through all of the variation models. If you finish with lesson time left, work through the models again to help further your understanding of these folds.

### → DISCUSS

What can you build when you put multiples of these models together?  
Can you think of any buildings or structures you have seen that may be based on this fold?

## WEEK 4 45m Sloyd: Level 7 - Lesson 28

### *Advanced Folding Techniques*

 Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

### → PRACTICE

Work through the models on stretch & skew p.044-049. You have five lessons to work through all of the variation models. If you finish with lesson time left, work through the models again to help further your understanding of these folds.

### → DISCUSS

What can you build when you put multiples of these models together?  
Can you think of any buildings or structures you have seen that may be based on this fold?

## WEEK 5 45m Sloyd: Level 7 - Lesson 29

### *Advanced Folding Techniques*

 Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

### → PRACTICE

Work through the models on stretch & skew p.044-049. You have five lessons to work through all of the variation models. If you finish with lesson time left, work through the models again to help further your understanding of these folds.

### → DISCUSS

What can you build when you put multiples of these models together?  
Can you think of any buildings or structures you have seen that may be based on this fold?



## Term 3

### WEEK 6 45m Sloyd: Level 7 - Lesson 30

#### *Advanced Folding Techniques*

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → READ

Read Polygons on p.051.

#### → PRACTICE

Look over the diagrams of polygons on p.051. Play around with working out the basic pleats in some of these new polygons.

### WEEK 7 45m Sloyd: Level 7 - Lesson 31

#### *Advanced Folding Techniques*

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → PRACTICE

For the remainder of Term 3, work through other projects throughout the rest of the book.

#### → NOTE

There are other videos that you can watch on Vimeo that will give examples of more projects in the book. Search for "Paul Jackson Folding Techniques" or "Paul Jackson Completes Pleats."

### WEEK 8 45m Sloyd: Level 7 - Lesson 32

#### *Advanced Folding Techniques*

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → PRACTICE

For the remainder of Term 3, work through other projects throughout the rest of the book.

### WEEK 9 45m Sloyd: Level 7 - Lesson 33

#### *Advanced Folding Techniques*

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

#### → PRACTICE

For the remainder of Term 3, work through other projects throughout the rest of the book.





## Term 3

WEEK 10

45m Sloyd: Level 7 - Lesson 34

Advanced Folding Techniques

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

→ **PRACTICE**

For the remainder of Term 3, work through other projects throughout the rest of the book.

WEEK 11

45m Sloyd: Level 7 - Lesson 35

Advanced Folding Techniques

Materials: Folding Techniques For Designers, A4 paper

PREP: Gather and prepare supplies for this model.

→ **PRACTICE**

For the remainder of Term 3, work through other projects throughout the rest of the book.

WEEK 12

45m Sloyd: Level 7 - Lesson 36

Exam Week

→ **EXAMS**

Answer question(s) related to course.

Questions will come from:

Folding Techniques For Designers.

# Sloyd: Level 7

## Examination

### Term 1

- Share one of the pieces you made this term with an adult and explain how you made it.
- Create a model from memory using the correct supplies to do so.

### Term 2

- Share one of the pieces you made this term with an adult and explain how you made it.
- Create a model from memory using the correct supplies.

### Term 3

- Share one of the pieces you made this term with an adult and explain how you made it.
- Create a model from memory using the correct supplies.